Geek Mind — One Agent Runtime, Multiple Embodiments

From perception to task plans, durable execution and verified recovery

Same agent, different bodies
Go2 · ARX X5 · simulation profiles
SHARED LOGIC · EMBODIMENT ADAPTERS

1 Perception
Sensors + World Understanding

See, hear and ground the current world

- Vision / RGB-D
- Sound / Speech
- Robot State
- Location · LiDAR · Depth
- VLM / Object Perception
- ASR / Language
- Platform & Joint State
- Spatial / 3D Grounding

Natural Language Data Bus + Structured World Observations
Scene · entities · robot state · source / timestamp · evidence references

2 Cortex + Runtime
Planning, Reasoning & Task Orchestration

Propose a plan. Execute verified progress.

Context Composer → LLM Planner
Goal · constraints · world · task state · budget · memory

Task Planner
Dedicated, configurable LLM client
Propose a finite, structured plan
Validate skills, arguments & goal

Task Plan / DAG
Nodes · dependencies · preconditions
Skill arguments · completion criteria
Verifier · resources · plan revision

Task Runtime
Schedule eligible nodes
Advance after independent verification
Pause · resume · cancel · revise

Persistent State & Memory — SHARED ACROSS THE LOOP · SQLITE

TaskStore	World State + Receipts	Memory Service
AUTHORITATIVE EXECUTION STATE	OBSERVATIONS & EVIDENCE	DERIVED KNOWLEDGE
Goal · plan versions · node states	Source · time · binding · freshness	Scoped facts · revisions · expiry
Budgets · outbox · events · controls	Command receipts · before / after	Verified experience → future context

Feedback
Update world and task state
Retrieve memory
Replan if needed

3 Skills / Tools
Contracts + Gateway

Call bounded skills with structured results

Skill Registry · MCP / Action APIs
- Navigation
- Manipulation
- Perception Tools
- Web / External APIs

Execution Gateway
Validate → persist intent → resource gates → dispatch
Stable command_id · timeout · query · cancellation
ActionRequest / ActionResult + explicit adapter capabilities

4 Robot Execution
Embodiment Adapters

Execute the skill. Return receipts and evidence.

Embodiment Adapter
Bind semantic skills to a platform
MCP · Zenoh · ROS 2 · native SDK

Controller / Policy
Movement · manipulation · interaction
Robot controllers / native policies

ActionResult + Evidence
command_id · status · effect
Native receipts + fresh observations

5 Verifier + Recovery
Evidence-based Progress

Check the outcome. Repair the remaining plan.

Node & Final Goal Verifiers
Completion criteria + fresh world evidence
Execution success is not goal completion

PASS	FAIL	INCONCLUSIVE
Advance / finish	Bounded replan	Hold / refresh evidence

Recovery Supervisor Same goal + remaining budget · UNKNOWN execution → query original command, reconcile before retry

EMBODIMENT · GO2
MOBILE EMBODIMENT



- UNITREE SDK · NATIVE / MID360 NAV
- D435 · LiDAR · ROBOT STATE
- LOCOMOTION · OPTIONAL ARM

EMBODIMENT · ARX X5
MANIPULATION EMBODIMENT



- ZENOH · GRASP CONNECTOR
- D435 RGB-D · JOINT / EEF STATE
- ARM · GRIPPER · GRASP / RESET

Implementation Scope

MAGIC v2 / OPTIMUS
Full plan → runtime → gateway path
Domain verification + durable state

GO2 / ARX X5 ADAPTATION
Existing hardware profiles retained.
New Gateway + domain-verifier integration remains to be completed.

Hardware v2 integration: planned
Core reference: MAGIC d917d39

Perceive → Plan → Persist → Execute → Verify → Update / Recover

Based on Geek-Mind + MAGIC d917d39 · Hardware: Unitree Robotics / TidyBot++ (CC BY 4.0)